| Program | B. Tech. (SoAE & SoCS) | Semester | I |
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| Course | Advanced Engineering Mathematics I | Course Code | MATH 1059 |
| Session | July-Dec 2025 | Unit III | Vector Calculus |

1. Find the rate of increase of the function $\phi(x, y, z) = xyz$ in the direction normal to the surface $x^2y + y^2x + z^2y = 3$ at point $(1,1,1)$.

2. Find the directional derivative of $\phi(x, y, z) = x^2 - y^2 + 2z^2$ at the point $P(1,2,3)$ in the direction of the line $PQ$ where $Q$ is the point $(5,0,4)$. In what direction it will be maximum ? Find the maximum value of it.

3. If $\theta$ is the acute angle between the surfaces $\phi_1(x, y, z) \equiv x^2 + y^2 + z^2 - 9 = 0$ and $\phi_2(x, y, z) \equiv z - x^2 - y^2 + 3 = 0$ at the point $P(2, 1, 2)$ then find the value of $\cos\theta$.

4. Find the domains $\mathcal{R}_1$ and $\mathcal{R}_2$ in the $3D$ −space such that $\nabla.\vec{F} > 0$ and $\nabla.\vec{F} < 0$ hold on $\mathcal{R}_1$ and $\mathcal{R}_2$ respectively for the vector field $\vec{F}(x, y, z) = (x^3)\hat{\imath} + (y^3)\hat{\jmath} + (z^3 - 3z)\hat{k}$.

5. Does there exist some $(a, b) \in \mathbb{R}^2$ such that the vector field
$$\vec{F}(x, y, z) = (ax - x^2)e^{-x}\hat{\imath} + \{(bx + 1)ye^{-x}\}\hat{\jmath} + (xe^{-x}z)\hat{k}$$
is solenoidal on $\mathbb{R}^3$? Justify your answer.

6. Determine the constants $a$ and $b$ if
$$\vec{F}(x, y, z) = (2xy + 3yz)\hat{\imath} + (x^2 + axz - 4z^2)\hat{\jmath} + (3xy + 2byz)\hat{k}$$
is irrotational.

7. Find the constant $\alpha$ such that the force field
$$\vec{F} = (e^x z - \alpha xy)\hat{\imath} + (1 - \alpha x^2)\hat{\jmath} + (e^x + \alpha z)\hat{k}$$
is a conservative field. Also, find the scalar potential $\phi(x, y, z)$ such that $\nabla.\vec{F} = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$.

8. Verify Green's theorem for $\oint_C \left[\left(\frac{1}{y}\right)dx + \left(\frac{1}{x}\right)dy\right]$, where $C$ is the boundary of the region bounded by the parabola $y = \sqrt{x}$, and the line $x = 1, x = 4, y = 1$.

**9.** Consider the vector field $\vec{F} = r^{\beta}(y\hat{\imath} - x\hat{\jmath})$ where $\beta \in \mathbb{R}, \vec{r} = x\hat{\imath} + y\hat{\jmath}$ and $r = |\vec{r}| = \sqrt{x^2 + y^2}$. If

$$\left| \oint_C \vec{F}.d\vec{r} \right| = 2\pi$$

along the circle $C: x^2 + y^2 = a^2$ oriented counter clockwise then find the value of $\beta$.

**10.** Determine constants a, b, c so that the force field

$$\vec{F} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$$

is conservative. Also find

a) The scalar potential $\emptyset$

b) find the work done in moving a particle under this force field from $(1,2 - 4)$ to $(3,3,2)$.

**11.** Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ where $\vec{F} = 18z\hat{\imath} - 12\hat{\jmath} + 3y\hat{k}$ and $S$ is the part of the plane $2x + 3y + 6z = 12$ in the first octant.